

Intersection Points

Time limit: 1.00 s

Memory limit: 512 MB

Given n horizontal and vertical line segments, your task is to calculate the number of their intersection points.

You can assume that no parallel line segments intersect, and no endpoint of a line segment is an intersection point.

Input

The first line has an integer n : the number of line segments.

Then there are n lines describing the line segments. Each line has four integers: x_1, y_1, x_2 and y_2 : a line segment begins at point (x_1, y_1) and ends at point (x_2, y_2) .

Output

Print the number of intersection points.

Constraints

- $1 \leq n \leq 10^5$
- $-10^6 \leq x_1 \leq x_2 \leq 10^6$
- $-10^6 \leq y_1 \leq y_2 \leq 10^6$
- $(x_1, y_1) \neq (x_2, y_2)$

Example

Input	Output
3 2 3 7 3 3 1 3 5 6 2 6 6	2